SENTIMENT ANALYSIS SERVICE

So tay ky thuat va bao cao phan quyen so huu

Tai lieu handover, readiness review va phan cong ownership

Tech Lead	Le Quang Tuyen
ML Engineering Lead	Duong Binh An
AI Engineering Lead	Do Quoc Trung
Solution Engineering Lead	Duong Hong Quan
Backend Engineering Lead	Pham Duc Long
UI/UX Engineering Lead	Nguyen Le Hong Nhi

Nguyen tac nguon su that: tai lieu nay uu tien implementation trong repository, sau do den cau hinh runtime, tests, va cuoi cung moi den cac bao cao cu. Neu code va tai lieu mau thuan, van hanh se tin code.

Contents

1 Danh gia dieu hanh

1.1 Muc dich kinh doanh

San pham bien feedback van ban thanh tin hieu van hanh co cau truc: sentiment tong the, sentiment theo aspect, dau hieu sarcasm, va giai thich muc token. Gia tri kinh doanh nam o phan tich review, theo doi chat luong dich vu, ho tro triage support, va demo explainable AI.

1.2 Muc dich san pham

Repository dang dong thoi chua hai san pham:

- dich vu inference online qua Angular va FastAPI
- he thong ML offline cho preprocessing, finetuning, evaluation, MLflow, va ONNX export

1.3 Muc dich ky thuat

Codebase the hien duoc day du chuoi phan phoi ML:

- DVC quan ly data flow
- Hugging Face va PEFT cho training
- ONNX runtime cho serving
- Prometheus va Grafana cho observability
- Docker Compose cho dong goi va chay he thong

1.4 Gia tri cot loi

Gia tri chinh khong nam o model moi la. Gia tri nam o viec dong goi sentiment inference thanh mot dich vu co explainability, batch processing, monitoring, va data/model lineage ro rang.

1.5 Danh gia do truong thanh

Moc	Danh gia
Prototype	Da vuot qua. Repo co API contracts, tests, monitoring, Docker, DVC, MLflow, ONNX export.
MVP	Da vuot qua. Co predict, explain, health, metrics va batch CSV.
Advanced MVP	**Trang thai hien tai**. He thong da co hanh vi san pham end-to-end va vong doi ML khong don gian.
Pre-Production	Moi dat mot phan. Van con thieu auth, CORS dang mo, state trong memory, batch status mock, va chua co durable job orchestration.
Production Ready	Chua dat. Phu hop cho demo, noi bo, va pilot kiem soat chat; chua phu hop enterprise production.

1.6 Vi sao chua san sang production

- `src/main.py` mo toan bo endpoint inference va evaluation ma khong co auth.
- `/batch_status/{job_id}` la mock, khong co queue hay persistent state.
- frontend cho phep chon ngon ngu nhung `SentimentAnalysisService.predict()` bo qua truong `lang`.
- Grafana duoc khoi tao voi mat khau mac dinh `admin`.
- health check chi xac nhan model da load, khong xac minh toan bo dependency xuong duoi.

2 Mo hinh tinh than he thong

2.1 Business view

```
flowchart LR
  Customer[Khach hang hoac Analyst]
  UI[Angular UI]
  API[FastAPI Service]
  Insight[Sentiment va Aspect Insight]
  Ops[Monitoring va Reports]
  Improve[Retraining va Export]
  Customer --> UI --> API --> Insight --> Ops --> Improve
```

2.2 Product view

```
flowchart TB
  A[Single Predict]
  B[Explain SHAP]
  C[Batch CSV Predict]
  D[Background Evaluate]
  E[Dashboard va Metrics]
  UI[Frontend]
  UI --> A
  UI --> B
  UI --> C
  API[Backend]
  A --> API
  B --> API
  C --> API
  D --> API
  API --> E
```

2.3 Engineering view

```
flowchart LR
  FE[app/sentiment-analysis-chatbot]
  BE[src/main.py]
```

```
  MODEL[src/model]
  DATA[src/data]
  TRAIN[src/scripts + src/training]
  OBS[infra + src/monitoring]
  FE --> BE --> MODEL
  DATA --> TRAIN
  TRAIN --> MODEL
  BE --> OBS
```

2.4 Infrastructure view

```
flowchart LR
  Browser --> Nginx[Frontend container :80]
  Nginx --> FastAPI[fastapi_app :8000]
  FastAPI --> ML[Model runtime in-process]
  FastAPI --> Metrics[/metrics]
  Metrics --> Prometheus[:9091]
  Prometheus --> Grafana[:3000]
  FastAPI --> Eval[/evaluate]
  Eval --> MLflow[:5005]
```

2.5 AI/ML view

```
flowchart LR
  Raw[data/raw]
  Prep[src.data.pipeline.py]
  Validate[src.data.validators.py]
  Train[src.scripts.run_finetuning.py]
  Eval[src.scripts.evaluate_finetuned.py va src.model.evaluate.py]
  Export[src.scripts.export_onnx.py]
  Serve[src.model.baseline.py]
  Raw --> Prep --> Validate --> Train --> Eval --> Export --> Serve
```

3 Luong nghiep vu end-to-end

Buoc	**Muc tieu kinh doanh**	**Implementation ky thuat**	**Owner**
Customer	Gui text de duoc phan tich	Nguoi dung tuong tac voi chat input hoac CSV upload trong Angular components.	UI/UX Lead
UI	Thu thap text, voice transcription, hoac CSV	`ChatInputComponent` phat su kien text; `BatchUploadComponent` upload file multipart.	UI/UX Lead
API	Validate request va route dung execution path	`src/main.py` dung Pydantic schema trong `contracts/schemas.py`.	Backend Lead
Inference	Sinh sentiment, aspects, sarcasm, hoac SHAP	`BaselineModelInference` goi HF hoac ONNX runtime va zero-shot ABSA.	AI Lead
Result	Tra response co cau truc cho UI hoac caller	`PredictResponse`, `ExplainResponse`, va batch response.	Backend Lead

Buoc	Muc tieu kinh doanh	Implementation ky thuat	Owner
Monitoring	Ghi nhan request count va latency	middleware trong `src/monitoring/metrics.py` va endpoint `/metrics`.	Solution Lead
Feedback	Xem metrics va report evaluation	Grafana, MLflow, JSON reports trong `reports/` va `data/reports/`.	Solution Lead / ML Lead
Retraining	Nang cap model/adapters	`src/scripts/run_finetuning.py`, task configs, class weights, LoRA adapters.	ML Lead
Deployment	Dong goi artifact moi de phuc vu runtime	Dockerfiles, Compose, ONNX export scripts, mounted model dirs.	Solution Lead / Backend Lead

4 Kien truc chi tiet

4.1 Frontend

Trach nhiem: message flow, voice capture, luu lich su, explain trigger, batch upload, theme control.
Phu thuoc: Angular, HttpClient, local storage, backend `/api/*`.
Rui ro: contract drift, bug ngon ngu, frontend test it.
Ownership: UI/UX Lead primary, Backend Lead secondary.

4.2 Backend

Trach nhiem: startup lifecycle, dependency injection, validation, exception mapping, health, metrics, evaluate trigger, batch orchestration.
Phu thuoc: FastAPI, pandas, model interface, monitoring middleware.
Rui ro: single-process bottleneck, route ton tai nhung chi la mock, khong auth.
Ownership: Backend Lead primary, Solution Lead secondary.

4.3 Inference layer

Trach nhiem: sentiment scoring, sarcasm adapter, ABSA extraction, SHAP, ONNX load, language validation.
Phu thuoc: Hugging Face, PEFT, SHAP, onnxruntime, tokenizer va model artifacts.
Rui ro: startup latency, SHAP ton tai chi phi cao, artifact mismatch.
Ownership: AI Lead primary, ML Lead secondary.

4.4 Training layer

Trach nhiem: load dataset, dedup, split, xu ly imbalance, LoRA config, trainer, MLflow logging.
Phu thuoc: Datasets, Transformers, PEFT, sklearn, task configs.
Rui ro: data skew, phu thuoc file raw, reproducibility chi dung trong dung environment.
Ownership: ML Lead primary, AI Lead secondary.

4.5 Monitoring layer

Trach nhiem: request metrics, dashboard provisioning, alert rules.
Phu thuoc: Prometheus, Grafana, middleware instrumentation.
Rui ro: khong co tracing, chua co business SLI manh.
Ownership: Solution Lead primary, Backend Lead secondary.

4.6 Deployment layer

Trach nhiem: image build, service topology, ports, resource limits, network, mounted volumes.
Phu thuoc: Dockerfile, Dockerfile.train, Compose, model va data dirs.
Rui ro: secret handling yeu, default credentials, khong co rollback automation.
Ownership: Solution Lead primary, Backend Lead secondary.

5 Ban do ownership repository

Folder	Muc dich	Primary	Secondary	Business Impact	Modification Risk	Knowledge Risk
`app/`	Angular UI cho nguoi dung	UI/UX	Backend	High	Medium	Medium
`contracts/`	API va model contracts dung chung	Backend	AI	High	High	Medium
`data/`	Raw, processed, eval, report artifacts	ML	AI	High	High	High
`docs/`	Bao cao cu va handbook	Tech Lead	All leads	Medium	Low	Low
`infra/`	Prometheus va Grafana provisioning	Solution	Backend	High	Medium	Medium
`mlruns/`	Local MLflow store	ML	Solution	Medium	Low	Medium
`models/`	Adapters va ONNX artifacts	AI	ML	High	High	High
`notebooks/`	Colab/manual pipeline	ML	AI	Medium	Medium	High
`reports/`	Evaluation va benchmark outputs	ML	AI	Medium	Low	Medium
`src/data/`	Download, preprocess, validate	ML	AI	High	High	Medium
`src/model/`	Runtime inference va export logic	AI	ML	High	High	High
`src/monitoring/`	Prometheus instrumentation	Solution	Backend	Medium	Low	Low
`src/scripts/`	CLI entrypoints	ML	AI	High	Medium	Medium
`src/training/`	Training internals	ML	AI	High	Medium	Medium
`tests/`	Regression safety net	Tech Lead	Owning team	High	Medium	Medium

Folder	Muc dich	Primary	Secondary	Business Impact	Modification Risk	Knowledge Risk
`.github/`	CI behavior	Backend	Tech Lead	High	Medium	Medium

6 Trace thuc thi runtime

Flow	File / Class	Function	Input	Output	Exception Path	Owner
Startup	`src/main.py` / module	`lifespan()`	env `MODEL_MODE`	`ml_model` san sang	`ModelError` khi load loi	Backend + AI
Health	`src/main.py` / API	`health_check()`	HTTP GET	`HealthResponse`	503 neu model chua ready	Backend
Predict	`src/main.py` / API	`predict()`	`PredictRequest`	`PredictResponse`	400/500 neu lang hoac model loi	Backend
Predict core	`src/model/baseline.py` / `BaselineModelInference`	`predict_single()`	text, lang	`PredictionResult`	unsupported language hoac model failure	AI
Explain	`src/main.py` / API	`explain()`	`ExplainRequest`	`ExplainResponse`	explainer failure	Backend + AI
Batch Predict	`src/main.py` / API	`batch_predict()`	uploaded CSV	`BatchPredictResponse`	400 invalid CSV, 500 batch failure	Backend
Batch core	`src/model/baseline.py` / `BaselineModelInference`	`predict_batch()`	list text	list `PredictionResult`	invalid batch size	AI
Evaluate trigger	`src/main.py` / API	`run_evaluate()`	POST	status JSON	409 neu dang chay, state memory co the mat	Backend + ML
Evaluate status	`src/main.py` / API	`evaluate_status()`	GET	state JSON	stale sau restart	Backend
Offline eval	`src/model/evaluate.py`	`evaluate_on_dataset()`	processed test frame	metrics dict	tra ve error payload neu split rong	ML + AI

6.1 Sequence diagrams

Startup

```
sequenceDiagram
  participant U as Uvicorn/FastAPI
  participant A as src.main.lifespan
```

```
  participant C as ModelConfig
  participant M as BaselineModelInference
  participant O as OnnxInferenceSession hoac HF model
  U->>A: app startup
  A->>C: doc MODEL_MODE
  A->>M: khoi tao
  M->>O: load tokenizer/model/session
  A->>M: preload()
  M->>M: warm ABSA pipeline
  A-->>U: app ready
```

Predict

```
sequenceDiagram
  participant User
  participant FE as Angular UI
  participant API as FastAPI /predict
  participant LD as LanguageDetector
  participant MI as BaselineModelInference
  participant ABSA as Zero-shot pipeline
  User->>FE: nhap text
  FE->>API: POST /api/predict {text}
  API->>LD: resolve_request_language()
  API->>MI: predict_single(text, lang)
  MI->>MI: sentiment logits
  MI->>ABSA: aspect extraction
  MI-->>API: PredictionResult
  API-->>FE: PredictResponse
  FE-->>User: render ket qua
```

7 Team handbooks

7.1 7.1 Tech Lead Handbook

Owner: Le Quang Tuyen.
Responsibilities: kien truc, code-review bar, readiness approval, giai quyet xung dot cross-team.
Systems owned: kien truc toan he thong, standards, quality gates.
Operational duties: review readiness va chi huy incident Sev-1.
Release duties: approve contract changes va security exceptions.
Risks: architecture drift va ownership mo ho.
KPIs: escaped defects, release predictability, readiness score.

7.2 7.2 ML Engineering Handbook

Owner: Duong Binh An.
Responsibilities: datasets, preprocessing, validation thresholds, finetuning, evaluation, MLflow.
Systems owned: `src/data/`, `src/training/`, scripts training, `data/`, `reports/`.
Operational duties: dam bao dataset integrity va evaluation reproducibility.
Release duties: xac nhan metric truoc khi promote artifact.

Risks: knowledge silo o data lineage va imbalance logic.
KPIs: macro F1, per-language F1, validation pass rate.

7.3 7.3 AI Engineering Handbook

Owner: Do Quoc Trung.
Responsibilities: inference behavior, ONNX runtime, ABSA, sarcasm, SHAP, language detection.
Systems owned: `src/model/`, `models/`.
Operational duties: model load diagnostics, latency tuning, artifact compatibility.
Release duties: ky duyệt artifact integrity va inference regression.
Risks: artifact mismatch, SHAP chi phi cao, provider issues.
KPIs: p95 latency, load success, inference error rate.

7.4 7.4 Solution Engineering Handbook

Owner: Duong Hong Quan.
Responsibilities: integration end-to-end, monitoring stack, deployment topology.
Systems owned: `infra/`, Compose topology, release runbooks.
Operational duties: giu observability va environment consistency.
Release duties: verify service health, ports, volume, dashboards.
Risks: dependency drift, insecure defaults, nhieu buoc manual.
KPIs: deployment success, monitoring freshness, MTTR.

7.5 7.5 Backend Engineering Handbook

Owner: Pham Duc Long.
Responsibilities: API semantics, validation, exception handling, evaluate va batch endpoints.
Systems owned: `src/main.py`, `contracts/`, backend CI.
Operational duties: endpoint health, error triage, backward compatibility.
Release duties: giu OpenAPI dung va schema on dinh.
Risks: route drift, state memory, expensive call khong gioi han.
KPIs: 5xx rate, schema-change defect rate.

7.6 7.6 UI/UX Engineering Handbook

Owner: Nguyen Le Hong Nhi.
Responsibilities: user journey, interaction quality, explain visualization, batch UX.
Systems owned: Angular application.
Operational duties: validate client behavior voi backend contract.
Release duties: dam bao feature hien thi dung voi backend support that.
Risks: contract mismatch, test it, state local storage.
KPIs: task completion, UI defect rate, explain usage.

8 RACI matrix

Workstream	Tech	ML	AI	Solution	Backend	UI/UX
Feature development	A	C	C	C	R	R
Model training	C	A/R	C	I	I	I
Model deployment	A	C	R	R	C	I
API changes	A	I	C	C	R	C
Production release	A	C	C	R	R	C
Incident management	A	C	R	R	R	I
Monitoring	C	I	C	A/R	R	I
Security	A	I	C	R	R	C
Architecture changes	A/R	C	C	C	C	C

9 Data va model lineage

```
flowchart LR
  Raw[data/raw/*.csv]
  Validation1[src.data.downloader.validate_raw_schema]
  Prep[src.data.pipeline.PreprocessingPipeline.run]
  Validation2[src.data.validators.DataQualityValidator.validate]
  Train[src.scripts.run_finetuning.train]
  Eval[src.model.evaluate hoac src.scripts.evaluate_finetuned]
  Track[MLflow runs]
  Export[src.scripts.export_onnx]
  Runtime[src.main -> BaselineModelInference]
  Predict[PredictResponse]
  Raw --> Validation1 --> Prep --> Validation2 --> Train --> Eval --> Track
 --> Export --> Runtime --> Predict
```

10 Security review

Finding	Severity	Owner	Huong khac phuc
Khong co authentication/authorization cho inference va evaluation	Critical	Backend + Solution	them API auth, role-based access cho admin flows, enforce tai gateway
CORS dang mo `allow_origins=["*"]`	High	Backend	khoa origins theo environment
Grafana mac dinh password `admin`	High	Solution	dua secret ra ngoai Compose va bat buoc rotate
Build args chua secret DagsHub	High	Solution	chuyen sang runtime secret hoac CI secret mount
CSV upload chua co file-size/content scanning tot	Medium	Backend	them size limit, content-type check, parser hardening

Finding	Severity	Owner	Huong khac phuc
Public evaluation endpoint	High	Backend	dua vao admin surface hoac bo khoi public runtime
Khong co rate limiting	High	Backend + Solution	them control o proxy va application

11 Operations handbook

Tinh huong	Dau hieu	Nguyen nhan	Buoc debug	Khac phuc	Owner
API down	`/health` fail, UI loading mai	model load fail, crash, path sai	xem backend logs, check `MODEL_MODE`, verify model files	rollback artifact hoac image	Backend + AI
Model failure	500 tren predict/explain	tokenizer/session mismatch, artifact hong	reproduce API call, inspect `ModelError`	restore artifact tot cu	AI
Latency spike	p95 tang, Grafana alert	SHAP, ABSA, CPU saturation	tach predict va explain, xem CPU limits	giam heavy path hoac tang tai nguyen	AI + Solution
Monitoring failure	Grafana rong, scrape loi	metrics endpoint hoac config drift	query `/metrics`, inspect Prometheus targets	sua scrape target hoac datasource	Solution
Batch failure	400/500 khi upload CSV	malformed CSV, thieu cot `text`	reproduce voi sample CSV, doc parser error	fix input hoac harden parser	Backend
Deployment failure	Compose services unhealthy	config drift, missing volume, build loi	xem `docker-compose` logs	restore config, rebuild, revert	Solution

12 Technical debt register

Level	Debt Item	Impact	Risk	Owner	Fix Cost	Priority
Critical	Public unauthenticated inference va evaluation	enterprise blocking	security/abuse	Backend	Medium	P0
High	`/batch_status` la mock	semantics sai	operational confusion	Backend	Medium	P1
High	Language selector khong di den backend	user expectation sai	correctness drift	UI/UX	Low	P1
High	Grafana default credentials va CORS rong	security posture yeu	compromise risk	Solution	Low	P1
Medium	Evaluation state nam trong memory	continuity yeu	stale status	Backend	Low	P2
Medium	SHAP khong co guardrails ro	latency spike	runtime instability	AI	Medium	P2
Low	Frontend test coverage mong	regression detection cham	maintainability	UI/UX	Medium	P3

13 Rui ro to chuc

- **Knowledge silo:** inference internals va artifact promotion dang tap trung vao AI/ML.
- **Single point of failure:** `src/model/baseline.py`, `src/scripts/run_finetuning.py`, va `docker-compose.yml`.
- **Bus factor:** thap cho ONNX export, imbalance logic, observability provisioning.
- **Cross-team dependencies:** thay doi label set can ML, AI, Backend, UI, va tests cung di chuyen.
- **Ownership gaps:** chua co security owner ro rang va release manager chinh thuc.

14 Production readiness review

Area	Score	Status	Ly do
Architecture	74	Amber	layering tot nhung runtime va async semantics chua that
Security	32	Red	thieu auth, CORS rong, default credentials
Testing	76	Amber	Python tests tot, frontend tests mong
Monitoring	71	Amber	co Prometheus/Grafana nhung chua co tracing
Performance	63	Amber	ONNX co loi ich nhung SHAP/ABSA van dat do
Reliability	58	Amber	co health nhung khong co durable queue/state
ML lifecycle	79	Amber	DVC, MLflow, training, eval, export kha day du
Deployment	60	Amber	Compose hop cho env kiem soat, chua hop enterprise
Operations	55	Amber	runbook co the viet duoc nhung tu dong hoa it
Ownership	70	Amber	role kha ro sau handbook nhung van phu thuoc ca nhan
Overall	**64**	**Amber**	advanced MVP, chua duyet enterprise production

15 Roadmap

15.1 3 thang

- them authentication, rate limiting, CORS theo environment. Owner: Backend + Solution. Priority: P0.
- thay `/batch_status` bang durable job tracking. Owner: Backend. Priority: P1.
- sua frontend language propagation va dong bo supported-language UX. Owner: UI/UX + Backend. Priority: P1.

15.2 6 thang

- them artifact registry/checksum validation. Owner: AI + ML. Priority: P1.
- them distributed tracing va model SLIs. Owner: Solution. Priority: P2.

- tach admin evaluation khoi public inference deployment. Owner: Backend + Solution. Priority: P1.

15.3 12 thang

- chuyen khoi Compose-only sang managed deployment co secret va autoscaling. Owner: Solution. Priority: P1.
- formalize model registry, approval gate, rollback policy. Owner: Tech Lead + ML + AI. Priority: P1.
- mo rong multilingual support chi sau khi data, evaluation, UI dong bo that. Owner: ML + AI + UI/UX. Priority: P2.

16 Huong dan onboarding

Moc	Muc tieu onboarding
Day 1	Doc phan lien quan den vai tro: Backend doc `src/main.py` va `contracts/`; AI doc `src/model/`; ML doc `src/data/` va `params.yaml`; UI/UX doc `app/`; Solution doc Compose va infra.
Day 3	Chay tests lien quan, trace mot request end-to-end, ghi nhan mot discrepancy giua docs va code.
Day 7	Lam mot thay doi nho co review va cap nhat test.
Day 14	Tu minh chay mot workflow day du trong pham vi vai tro.
Day 30	Co the tu nhan mot issue van hanh trong vung so huu va review thay doi cua nguoi khac.

17 CTO review

17.1 Co phe duyet production launch khong?

Khong, neu la production mo hoac enterprise production. Co the phe duyet demo noi bo va pilot rat kiem soat sau khi hardening bao mat co ban.

17.2 Co phe duyet enterprise customers khong?

Khong. Security va operations controls hien tai chua dat muc toi thieu.

17.3 Co phe duyet scaling khong?

Chua. Kien truc co huong scale duoc, nhung deployment va control plane hien tai chua san sang.

17.4 Dieu dang lo nhat la gi?

Nguy hiem nhat la *false confidence*: repo nhin rat giong san pham production, nhung mot so control quan trong van thieu hoac chi la mock.

17.5 Nen dau tu tiep vao dau?

Uu tien hardening nen tang: authentication, rate limiting, artifact governance, va durable batch execution. Sau do moi dau tu manh hon cho multilingual va model operations.

Phu luc discrepancy

- README cho vi du health chi co `["en"]`, nhung `ModelConfig.supported_languages` la `("en","vi")`. Tac dong: tai lieu noi mot dang, backend support mot dang.
- README mo ta batch theo kieu async job, nhung `/batch_predict` thuc te la synchronous va `/batch_status` chi la mock. Tac dong: client co the tin vao polling semantics khong ton tai.
- Frontend hien nhieu ngon ngu trong `ChatInputComponent`, nhung backend chi support `en` va `vi`; service layer con bo qua explicit language. Tac dong: affordance vuot qua nang luc that.
- Compose dung `http://mlflow:5000` noi bo container, host expose `5005`, trong khi `params.yaml` default `http://localhost:5000`. Tac dong: assumptions khac nhau giua local CLI va container runtime.